4. GET

It is a method used to retrieve or read data from a server. Its purpose is to fetch information and read existing data.
Example: A weather app requesting today's weather information from a server.
Mozilla. (1998). HTTP GET method.
Available at: Mozilla HTTP GET Guide.

Question 22.

22. JSON ~~stands for~~ is a lightweight data ~~format~~ format used to store and exchange data between systems and applications. It organises data using key-value pairs, making it easy for both human and computers to read and understand. JSON is widely used in web applications, APIs, mobile apps and cloud services.
(ChartGPT)

JSON is commonly used because it is simple, lightweight and supported by many programming languages and systems. APIs use JSON to send and receive information between applications over the internet. JSON is easy for humans to read,
Example: When a weather app requests weather data from a server, the information is often returned in JSON format.
(ChartGPT)

Question 23

23. Prompt engineering is the process of designing and writing clear instructions or ~~re~~questions for AI to get accurate, useful, and high quality responses. It is commonly used with AI tools such as Claude and ChartGPT.

Prompt engineering focuses on improving the inputs so that the AI understands exactly what the user want after a prompt is given to an AI tool.
Google Cloud. (2026). What is Prompt Engineering?
Available at: Google Cloud Prompt Engineering Guide.

Prompt Engineering is important because the quality of the AI's response depends heavily on the quality of the prompt. Clear and detailed prompts help AI systems produce more accurate, relevant and useful answers. It reduces misunderstandings, It saves time.
(Chart GPT)

Examples of poorly written and improved Prompts.

1. Poor prompt.
Example: Tell me about databases.
Improved prompt.
Example: Explain what is a database, describe its importance in business and provide three real-world examples using simple language suitable for a first-year student.

An improved prompt is better because it clearly explains the topic, the level of detail required, the target audience as well as specific information needed.
Microsoft (2026.)-Prompt Design Concepts.
Available at: Microsoft Prompt Engineering Overview.

Question 24.

24. MAP function is a programming function used to apply a specific operation or transformation to every item in a list, array or collection of data.

MAP creates a new collection containing the updated values without changing the original data. It is commonly used in programming & data processing because it allows developers to process large amounts of data efficiently with less code.
(ChartGPT)

In programming and data processing, the MAP function takes a collection of data such as list or array. It also takes a function or operation to apply to each item. MAP function then processes each item one by one and applies the operation. After it then returns a new collection with the tranformed values.

Simple example in Python.

Numbers = (1, 2, 3, 4)
using map to multiply each number by 2
results = List (map (lambda x: x * 2, numbers)

Output

[2, 4, 6, 8]

Explanation.

The original list is (1,2,3,4). The map ( ) function applies x * 2 to every number. A new list is with doubled values is created.

Question 25

Question 25.

25. Cloud Computing is the delivery of computing services such as serves, storage, databases, networking and software over the internet instead of using local computers or physical serves. It allow users and businesses to access technology resources on demand and from anywhere with an internet connection.
IBM. (2026.). What is cloud Computing?
Available at: IBM cloud Computing Guide.

Main cloud services models.

1. Infrastucture as a service (Iaas)
It provides basic computing infrastructure as a virtual servers, storage and networking over the internet. Users manage the operating systems and applications while the cloud provider manages the physical hardware.
Amazon Web Services. (2026). What is Iaas?
Available at: AWS Iaas Overview.

2. Platform as a service (Paas)
It provides a platform where developers can build, test, and deploy applications without managing the underlying hardware or operating systems.
Example: Google App Engine allows developers to create and deploy applications on Google's cloud platform.
Google Cloud. (2026). What is Paas?
Available at: Google Cloud PaaS Guide.

3. Software as a service (saas)
It provides ready-to-use software applications over the internet. Users access the software through a web browser without installing or managing it locally.

Example: Microsoft 365 provides online tools such as Word, Excel, Outlook, and Teams through the cloud.
Microsoft. (2026). What is SaaS?
Available at: Microsoft SaaS Overview.

Question 26.

26. A version Control system is a tool that helps users track and manage changes made to files, document, or code over time. It allows multiple people to work on the same project while keeping a history of all changes. It helps users to track file changes, restore previous versions, collaborate with teams, prevent accident data loss. VCS are commonly used in software development, data analysis and collaborative projects.
Atlassian. (2026). What is Version Control?
Available at: Atlassian Version Control Guide.

Git is a distributed version control system used to track changes in ~~life o~~ files and manage collaborative projects. It was designed to help developers work on projects efficiently while maintaining a complete history of changes. Git allows users to save versions of their work, collaborate with others, and safely experiment with changes without affecting the main project.

Git is widely used by data analysts and developers because it tracks project history. It supports team work and collaboration. Git allows easy backup and recovery. Git helps manage code and data analysis projects. Data analysts use Git to manage scripts, notebooks, and datasets, while developers use it to manage

software code.
Git. (2026). About Git.
Available at: Git official website.

A commit in Git is saved snapshot of changes made to a project. Each commit records what changes were made, who made them, and when they were made. Commits allow users to save progress, track project history, return to earlier versions if needed.
Example: A developer commits changes after fixing a bug or adding a new feature.
GitHub. (2026). Git Commit Basics.
Available at: GitHub Git Handbook.

A branch is a separate version or copy of a project where users can work on changes without affecting the main project. Branches are useful for fixing bugs, testing new features, experimenting safely.
Example: A team creates a new branch to develop a new feature while the main version of the software continues running normally.
Atlassian (2026). Git Branching.
Available at: Atlassian Git Branch Guide.

Question 27.
27. A Jupyter Notebook is an interactive web-based application that allows users to write and run code, display visualisations, add explanations, and document their work in one place. It combines code, text, charts and outputs into a single ~~document~~ document called a notebook. Jupyter Notebooks are widely used for data analysis, machine learning, research and education.

Jupyter Notebook most commonly used with the programming language Python because Python is widely used in data science, machine learning, and analytics. However, Jupter also supports other languages such as R, Julia and Scala through special kernels.
Python Software Foundation. (2026). Python Documentation.
Available at: Python Official Website.

Jupyter Notebook is popular among Data Analysts and Data Scientists because it makes it easy to write code, test ideas, analyse data and present results in one environment.
Example: A data analyst can clean data, create graphs, and explain findings all within one notebook.
IBM. (2026). Jupyter Notebooks Explained.
Available at: IBM Jupyter Notebook Guide.

Question 28

28. Python is a high-level, general-purpose programming language that is known for being simple, readable, and easy to learn. It is widely used for web development, automation software development, data analysis, artificial intelligence AI and machine learning.
Python Software Foundation. (2026.). About Python.
Available at: Python Offical website.

Python is popular for Data Analytics and AI because it is easy to learn, flexible and supported by many powerful libraries and tools. Python is compatible with tools such as Jupyter Notebook. Python allows data analysts and data scientists to process data, create

visualisations, build machine learning models and automate tasks efficiently.
IBM. (2020). Why Python for Data Science?.
Available at: IBM Python Guide.

Python Libraries Commonly used in Data Work.

1. Pandas.
It is a library used for data manipulation and analysis. It helps users organise, clean, filter and analyse structured data using tables called DataFrames. Panda is used for cleaning datasets, working with rows and columns, analysing spreadsheets.
(ChatGPT)

2. NumPy
Is a library used for numerical and mathematical operations in Python. It provides support for arrays, matrices and fast calculations. NumPy is used for mathematical calculations, scientific computing, working with large numerical datasets.
NumPy Developers (2025). NumPy Documentation.
Available at: Numpy Official Website.

3. Scikit-Learn
Scikit-learn is a machine learning library used to build predictive models and perform data analysis. Scikit-learn is used for data mining and analytics, machine learning algorithms.

4. Matplotlib.
Is a library used to create charts, graphs and data visualisations. It is used for line graphs, bar graphs, data visualisation and reporting.